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Bayesian Neural Network-based Uncertainty Quantification of Thermo-mechanical Coupling in Heat Pipe Cooled Reactors

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1. Introduction

Heat Pipe Cooled Reactors (HPRs), as a promising type of advanced Small Modular Reactor (SMR), have demonstrated unique advantages for remote power generation and space applications due to their passive heat removal capability, inherent safety features, and compact core design [1]. However, such compactness also introduces pronounced thermo-mechanical coupling characteristics. For instance, a strong interdependence exists among fuel heat generation, heat pipe conduction, and monolith thermal stress, where minor fluctuations in material properties or geometric parameters may be amplified into significant safety concerns. Previous studies have indicated that structural stress often represents the primary safety concern in HPR operation [2]. Yet systematic uncertainty quantification is still lacking, leaving a gap in linking material fluctuations to reactor-level safety margins, leaving a critical gap in linking material property fluctuations to reactor-level safety margins. Accordingly, this study places particular emphasis on stress responses under thermo-mechanical coupling and their associated uncertainty sources.

To address these challenges, this paper develops a Bayesian Neural Network (BNN) based surrogate modeling and uncertainty quantification framework that integrates both forward and inverse analyses. Within this framework, we constructed and validated an efficient BNN surrogate model tailored for thermo-mechanical coupling in HPRs, demonstrating its superior performance in both predictive accuracy and computational efficiency compared to traditional approaches. Building on this foundation, global sensitivity analysis with Sobol indices was employed to identify the most influential input parameters, and rapid Monte Carlo sampling was used to explore the overall uncertainty bounds of system responses. Moreover, by leveraging the posterior sampling capability of BNNs in combination with a Markov Chain Monte Carlo (MCMC) approach, we conducted inverse inference to deduce feasible ranges of critical input parameters from predefined output safety thresholds. Collectively, these contributions not only fill a key gap in the treatment of parameter uncertainties but also provide quantitative and actionable guidance for HPR design and optimization.

The remainder of this paper is organized as follows: Section 2 introduces the HPR thermo- mechanical model together with the BNN methodology, including the identification of uncertainty sources, the surrogate architecture, and the adopted UQ approaches. Section 3 presents the results and discussion, covering BNN performance evaluation, global sensitivity analysis, and inverse uncertainty inference. Finally, Section 4 concludes the paper with a summary of the main findings and an outlook on future research directions.

2. Methodology

2.1 Thermo-Mechanical Model and Uncertainty Sources

To enable subsequent BNN-based training and sensitivity analysis, this study first establishes a high-fidelity numerical model based on FEniCS [3] to simulate the thermo-mechanical coupling within the HPR core. Heat generated in the fuel is conducted through the SS316 monolith to the heat pipes and subsequently removed, while thermal expansion in the monolith induces stress fields.

The uncertainty sources are derived from empirical formulations of SS316 material properties. As a high-temperature structural material, the thermal conductivity, Young's modulus, Poisson's ratio, and thermal expansion coefficient of SS316 directly govern heat transfer efficiency and structural deformation. Prior studies have shown that certain parameters deviate from reference databases, hence validation and adjustment are necessary before practical application. Thus, the coefficients within these formulas are considered as key uncertainty sources. Since structural stress constitutes the primary safety concern in compact HPRs, and the material properties of SS316 directly govern this stress response, the coefficients defining these empirical properties are selected as the key uncertainty sources, thereby filling a critical gap in linking material fluctuations to structural safety margins. Eight parameters are included, covering the slope and intercept of Young's modulus, Poisson's ratio, the base and slope of thermal expansion, and the reference temperature, reference value, and slope of thermal conductivity. The governing equations of SS316 material properties are summarized as follows, while the corresponding parameters are detailed in Table I [4].

$$E(T) = (E_{slope} \cdot T + E_{int}) \cdot 10^{-6} \tag{1}$$

$$\alpha(T) = \alpha_{slope} \cdot T + \alpha_{base} \tag{2}$$

$$k(T) = ((T - T_{ref}) \cdot k_{slope} + k_{ref}) \cdot 1/100 \tag{3}$$

$$\nu = 0.31 \tag{4}$$

Table I. BNN Input Parameters and Perturbation Scheme.

Parameter	Symbol	Value	Perturbation（σ）
Young's modulus slope	E_{slope}	-7×10^{7} (Pa/K)	10%
Young's modulus intercept	E_{int}	2×10^{11} (Pa)	10%
Thermal expansion slope	α_{slope}	5×10^{-9} (K^{-2})	10%
Thermal expansion base	α_{base}	1×10^{-5} (K^{-1})	10%
Conductivity slope	k_{slope}	1/75 (W/m·K²)	10%
Reference conductivity	k_{ref}	23.2 (W/m· K)	10%
Reference temperature	T_{ref}	923.15 (K)	10%
Poisson's ratio	ν	0.31	10%

The model outputs six indicators relevant to HPR safety and performance, including average and peak fuel temperature, average and peak monolith temperature, maximum global stress, and the radial coordinate of the outer wall (Table 2). These input–output variables constitute the basis for subsequent BNN training and sensitivity analysis.

Table II. BNN Output Parameters.

Output Parameter	Symbol
Average fuel temperature	$T_{fuel,avg}$
Peak fuel temperature	$T_{fuel,max}$
Average monolith temperature	$T_{mono,avg}$
Peak monolith temperature	$T_{mono,max}$
Peak global stress	σ_{max}
Monolith radial border coordinate	$x_{mono,radial}$

2.2 Bayesian Neural Network (BNN) Architecture and Training

To address the computational bottlenecks of traditional simulations in uncertainty quantification, we developed an efficient surrogate model based on Bayesian Neural Networks (BNNs) [5]. The model architecture consists of an input projection layer, two Bayesian residual blocks, and an uncertainty output head. The input layer maps the eight uncertain parameters into a high-dimensional latent space, followed by two residual blocks for deep feature extraction. Each residual block incorporates Linear Reparameterization layers, which model weight distributions instead of fixed-point estimates, thereby enabling the natural representation of epistemic uncertainty. To ensure stable optimization of the deep network, GELU activation functions and LayerNorm layers are employed throughout the architecture.

At the output stage, the uncertainty head is composed of two branches, one predicting the mean response and the other the logarithmic variance. This bifurcated design allows the model not only to provide accurate predictions but also to yield uncertainty estimates that can be further leveraged in sensitivity analysis and inverse inference tasks.

The training dataset is generated by the FENICS numerical model, consisting of 10000 sets of input-output sample pairs, and is divided into training, validation, and testing sets in a ratio of 7:2:1. The model training adopts Adam optimizer and adaptive learning rate scheduling. Bayesian optimization is used to further adjust hyperparameters to improve the overall generalization performance and stability of the model. The loss function design introduces the Evidence Lower Bound (ELBO) to improve the credibility of uncertainty estimation while balancing prediction accuracy.

2.3 Uncertainty Analysis Methods

The proposed UQ framework integrates Sobol global sensitivity analysis and MCMC-based inverse inference, thereby enabling a systematic characterization of the thermo-mechanical behavior of HPRs. In the sensitivity analysis stage, the Sobol method is employed, whose theoretical foundation lies in variance decomposition: the total output variance can be decomposed into the independent contributions of each input parameter and their interactions [6]. By generating parameter samples using Saltelli sampling which ensures efficient coverage of the input domain—and evaluating them through the BNN surrogate, sensitivity indices can be obtained at a fraction of the computational cost required by direct simulations. The total-effect index (S_T) is of primary interest, as it quantifies the overall contribution of each input to output variance, including both direct and interactive effects, thereby offering quantitative insights into thermo-mechanical coupling mechanisms.

The inverse inference uses Markov Chain Monte Carlo (MCMC), which represents the most critical component of the framework [7]. Rooted in Bayesian inference, the method defines the posterior distribution of input parameters as：

$$p(\theta, y) \propto L(y|\theta)\cdot\pi(\theta) \tag{5}$$

where $\pi(\theta)$ denotes the prior distribution of inputs, $L(y|\theta)$ is the likelihood representing the agreement between BNN predictions and prescribed safety thresholds, and y denotes the target outputs satisfying safety requirements. By sampling from this posterior, MCMC explores the parameter space and identifies feasible regions of input parameters that ensure compliance with safety constraints. Compared with conventional margin-based deterministic designs, this

approach explicitly quantifies how parameter uncertainties translate into safety requirements, thereby yielding actionable ranges for parameter control. In practice, this enables engineers to distinguish parameters that require strict regulation from those that allow relaxed tolerances, achieving an optimal balance between safety assurance, manufacturing feasibility, and cost efficiency.

3. Results and Discussion

3.1 BNN Performance and Uncertainty Calibration

The BNN's performance on the test set shows low prediction errors across six outputs, with an average MAE of 0.6739 and an average MAPE of 0.8210. For uncertainty quantification, the average Prediction Interval Coverage Probability (PICP), computed using the 95% confidence interval (mean ± 1.96σ), reaches 0.9773, demonstrating that the predicted intervals reliably capture the ground truth values. Moreover, the BNN surrogate achieves a dramatic efficiency gain, reducing the runtime per evaluation from ~15 minutes with the high-fidelity FEniCS model to only ~0.5 seconds, thereby enabling rapid large-scale UQ analyses. Although the Expected Calibration Error (ECE) is relatively high at 0.4869, this mainly reflects calibration limitations and will be addressed in future work through post-hoc calibration strategies and further dataset/hyperparameter refinement.

In particular, the network employs Bayesian residual blocks with Linear Reparameterization layers, which model weight distributions instead of fixed-point estimates, thereby enabling a natural representation of epistemic uncertainty. This intrinsic capability is essential for assessing the model's reliability under extrapolative scenarios: when input parameters deviate significantly from the training domain, the predicted variance increases sharply, clearly signaling the boundaries of the model's learned knowledge.

3.2 Global Sensitivity Analysis

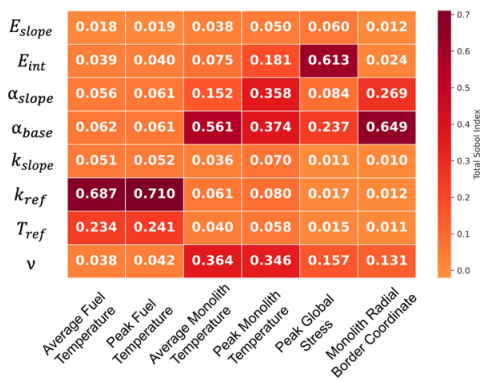

Fig. 1. Total Sobol indices of Sensitivity Analysis.

Sobol global sensitivity analysis reveals significant differences in how various HPR output indicators respond to parameter uncertainties, as is shown in Fig. 1.

For fuel temperature outputs, the thermal conductivity reference value k_{ref} is the most influential parameter, with total-effect indices (S_T) of 0.687 and 0.710, respectively—far exceeding others. This aligns with physical intuition: thermal conductivity directly governs heat transfer efficiency from fuel to heat pipes, dominating thermal performance.

For monolith temperature outputs, the thermal expansion base α_{base} and Poisson's ratio ν show greater sensitivity: α_{base} controls thermal strain from temperature rise, while ν regulates strain redistribution in the matrix. Together, they affect temperature distribution and thermal stress accumulation, indicating uncertainties in thermal expansion and elastic constants significantly impact temperature uniformity and local hot spot formation.

For monolith stress outputs, Young's modulus intercept E_{int} and thermal expansion slope α_{slope} are more sensitive: E_{int} reflects SS316's intrinsic stiffness , amplifying stress under given strain, and α_{slope} determines thermal strain's temperature-dependent rate , making stress highly sensitive to local thermal gradients. This shows structural integrity under thermo-mechanical coupling is vulnerable to uncertainties in stiffness and temperature-dependent expansion.

For the monolith radial border coordinate sensitivity is dominated by α_{base} (with an S_T value of 0.649), reflecting thermal expansion's cumulative effect on boundary shifts. Additionally, α_{slope} has non-negligible sensitivity for some structural outputs, indicating moderate fluctuations in temperature-dependent expansion rate significantly affect geometric boundary stability.

In summary, Sobol analysis clearly illustrates the thermos-mechanical coupling mechanism of HPR: thermal performance is primarily controlled by k_{ref}, structural responses are more sensitive to α -related parameters, and geometric boundary shifts are closely linked to α_{base}. These conclusions not only provide quantitative support for the thermo-mechanical coupling mechanism of HPR but also guide the prioritization of key uncertainty sources during the design phase.

3.3 Inverse Uncertainty Analysis

Inverse uncertainty analysis represents the core of this study, enabling the transformation of safety requirements into design specifications. Based on the sensitivity analysis results, three key parameters were selected for inversion: two parameters strongly sensitive to stress responses (the intercept of Young's modulus E_{int} and the base value of the thermal expansion coefficient α_{base}), and one parameter related to thermal performance (the reference thermal conductivity k_{ref}). Other parameters were kept fixed at their design values.

The safety-related target was imposed as constraints: maximum global stress σ_{max}, which must not exceed 45 MPa to ensure structural integrity of the SS316 monolith. Although the target is adopted here for demonstration, the framework can be flexibly extended to other safety

standards and supports the combination of multiple constraints.

The inverse analysis was carried out by coupling the Bayesian Neural Network (BNN) with Markov Chain Monte Carlo (MCMC) sampling. The likelihood function was defined using a truncated Gaussian, such that parameter sets violating the safety constraints were assigned negligible probability, ensuring that only feasible solutions were retained as posterior samples. This mechanism allows the feasible domain to be identified directly in the high-dimensional parameter space without additional forward evaluations.

To ensure the reliability of posterior distributions, 32 walkers were run for 6000 iterations, with 3000 burn-in steps discarded and a thinning factor of 3 applied to reduce autocorrelation. Convergence of all chains was confirmed via the trace plots (Fig. 2): after the 3000-step burn-in (marked by the red dashed line), the chains for all three inverted parameters exhibit stable, random fluctuations within consistent ranges (without directional drifts), and multiple walkers (colored lines) mix thoroughly. This indicates the MCMC sampling has sufficiently explored the posterior distribution and reached convergence. Ultimately, 32000 effective posterior samples were obtained, fully characterizing the feasible ranges of the inverted parameters.

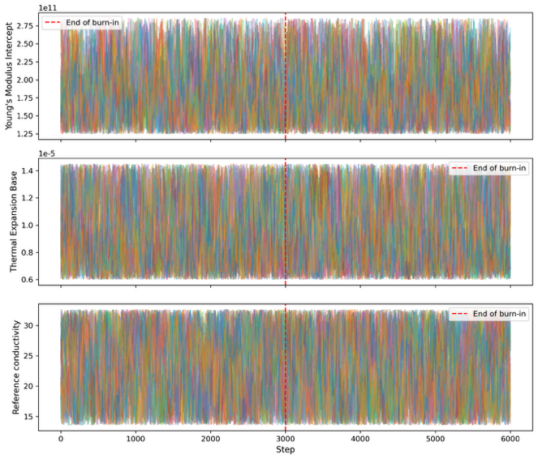

Fig. 2. MCMC Chain Convergence Trace Plots for Inferred Parameters.

The marginal and joint posterior distributions of the three inverted parameters exhibit strong non-Gaussian features and clear parameter compensation effects. E_{int} is skewed toward lower values, suggesting that reducing stiffness is an effective way to satisfy the stress constraint ($\sigma_{max} \leq 45$ MPa). α_{base} is also concentrated at the lower end and strongly correlated with E_{int}: feasible solutions cluster where E_{int} and α_{base} are simultaneously small, forming a compensation mechanism for stable stress responses. The distribution of k_{ref} is relatively flat, reflecting its lower sensitivity, but still shows strong interactions. Overall, these patterns reflect the multiplicative form $\sigma \propto E \cdot \alpha \cdot T$, where different parameter combinations can compensate each other to satisfy the stress threshold.

For this case, the conclusions are:

- E_{int} and α_{base}, as the most stress-sensitive parameters, are the primary factors requiring strict control.
- If achieving a low E_{int} or α_{base} is impractical, stricter limits on k_{ref} become necessary.
- To further constrain the feasible domain of inverted parameters, additional thermal constraints should be incorporated into the inverse analysis.

The key contribution of this study is that it translates the stress safety threshold ($\sigma_{max} \leq 45$ MPa) into engineering tolerance ranges for E_{int}, α_{base}, and k_{ref}, providing actionable quantitative guidance for HPR design.

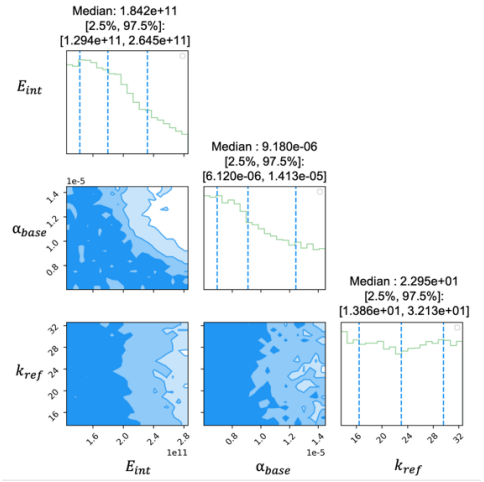

Fig. 3. The marginal posterior distributions of the three inverted parameters.

4. Conclusions

This paper developed a BNN-based framework for thermo-mechanical coupling uncertainty quantification of HPRs, incorporating both forward predictions and inverse inference. By leveraging high-fidelity numerical simulations for training, the BNN surrogate significantly reduced computational costs while simultaneously providing accurate predictions and reliable uncertainty estimates. Sobol global sensitivity analysis highlighted the dominant roles of thermal conductivity and thermal expansion in governing thermo-mechanical behavior, confirming their critical importance in HPR design. Building upon this, the inverse uncertainty analysis transformed safety requirements into quantitative parameter specifications, delineating feasible ranges for key inputs and thereby offering actionable guidance for engineering applications.

In summary, this study not only demonstrates the potential of BNNs for modeling complex thermo-mechanical coupling systems but also provides a novel pathway for design and optimization of advanced small modular reactors. Future work will focus on expanding the training dataset, improving uncertainty calibration through post-processing techniques, and extending the inverse framework to multi-objective constraints to further enhance engineering applicability and robustness.

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